Numerical Investigation of Rayleigh–Taylor Instability

VOF-style phase-fraction transport for a perturbed heavy-over-light interface

1. Aim

The purpose of this study is to examine the development of Rayleigh–Taylor instability when a heavier fluid is initially placed above a lighter fluid. The evolution of the perturbed interface is tracked while assessing phase-fraction boundedness and conservation for different Atwood numbers.

2. Computational Model and Parameters

The calculation is performed in a two-dimensional Cartesian domain. A projection-based flow solver is coupled with conservative upwind transport of the phase indicator. The initial interface is perturbed, and the resulting downward motion of the heavy-fluid spike and upward motion of the light-fluid bubbles are monitored.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx × ny	96 × 192
Kinematic viscosity, nu	0.0008
Final simulation time	2
Atwood numbers	0.2, 0.5 and 0.8
Gravitational acceleration	1
Poisson iterations	60

3. Governing Model and Boundary Conditions

The phase indicator alpha is transported through the computational domain to distinguish the two fluids. The momentum equation includes a buoyancy contribution proportional to the Atwood number. The source report describes a no-slip, impermeable enclosure, with the heavier fluid initially located above the lighter fluid and the interface given a small perturbation.

4. Numerical Procedure

The velocity field is advanced using a projection flow solver. Phase fraction is transported using a conservative upwind method. Clipping and conservative redistribution are then applied to maintain the bounded range $0 \leq \text{alpha} \leq 1$ while preserving the total phase fraction.

5. Quantities Monitored

The reported diagnostics include spike and bubble positions, mixing-layer width, divergence, relative mass error, minimum and maximum phase fraction, and the shape of the evolving interface. These quantities are compared for Atwood numbers of 0.2, 0.5 and 0.8.

6. Comparative Results

The table below summarizes the values reported at the final time. The Atwood-number-0.8 interface metrics are affected by jumps in the bubble-position and mixing-width measurements; those values are identified as artefacts in the source report.

Measured quantity	A = 0.2	A = 0.5	A = 0.8
Initial spike / bubble position (y)	0.48 / 0.52	0.48 / 0.52	0.48 / 0.52
Spike position at t = 2	0.398	0.319	0.235
Spike descent	0.084	0.163	0.247
Bubble position at t = 2	0.534	0.592	Unreliable after t ≈ 1.5
Mixing width at t = 0	0.037	0.037	0.037
Mixing width at t = 2	0.136	0.272	Affected by detection artefact
Net mixing-width growth	0.099	0.235	≈ 0.20 by t ≈ 1.5
Maximum absolute relative mass error	6 x 10^(-15)	4 x 10^(-15)	1.3 x 10^(-15)
Jumps in bubble / width history	None	None	t ≈ 1.5–1.65

The source table contains three highlighted entries for the A = 0.8 phase-fraction range, heavy-fluid fraction near the top wall, and interface shape, each marked for insertion from the field plot. These entries have not been estimated or filled in here.

7. Discussion

The plots show that the instability develops more rapidly as the Atwood number increases. The heavy-fluid spike moves downward at the centre of the domain, while the lighter fluid rises near the side walls. At t = 2, the spike position decreases from approximately 0.398 at A = 0.2 to 0.319 at A = 0.5 and 0.235 at A = 0.8.

The mixing-layer width also grows with time. For A = 0.2 it increases from about 0.037 to 0.136, while for A = 0.5 it reaches approximately 0.272 by t = 2. The A = 0.8 curve rises more sharply, but its bubble-position and mixing-width measurements exhibit repeated jumps between t ≈ 1.5 and 1.65. The source attributes these jumps to the interface-detection method responding to a diluted upper layer, rather than to actual bubble rise. The A = 0.8 interface metrics are therefore considered reliable only up to about t = 1.5.

The interface contours provide a visual comparison of the nonlinear development. At A = 0.2, the interface remains relatively smooth and retains a single-mode shape. At A = 0.5, the spike becomes rounded and the rising fluid near the side walls develops curled tips. The source identifies this as the beginning of nonlinear roll-up, which becomes stronger as the Atwood number increases.

8. Conservation and Phase-Fraction Boundedness

The volume-fraction mass error remains close to machine precision for all three Atwood numbers, with maximum reported values of approximately 6 x 10^(-15), 4 x 10^(-15) and 1.3 x 10^(-15) for A = 0.2, 0.5 and 0.8, respectively. The field plots indicate that the phase fraction remains within the prescribed interval from zero to one.

However, conservation and boundedness do not by themselves establish that the transported field is physically accurate. The source notes a gradual reduction in heavy-fluid fraction near the top wall, reaching about 0.8 for A = 0.2 and 0.6 for A = 0.5 at t = 2, even though that region should remain filled with the heavy fluid. Residual velocity divergence and/or the clipping and redistribution procedure may contribute to this depletion.

9. Conclusion

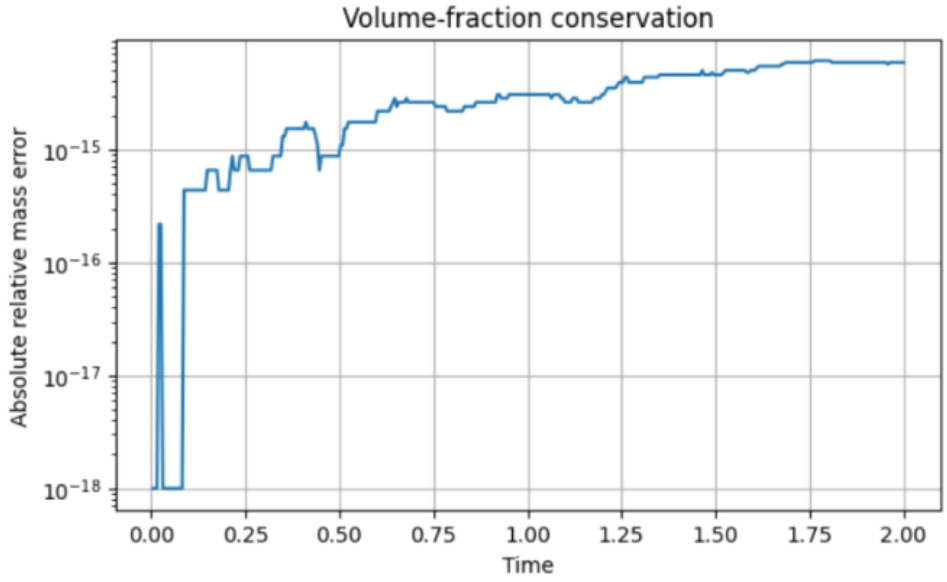
The numerical model reproduces the expected trend of stronger Rayleigh–Taylor growth at larger Atwood numbers. Increasing the Atwood number leads to greater spike descent, wider mixing layers and more pronounced interface deformation. The volume-fraction transport remains conservative to near round-off precision in the reported cases.

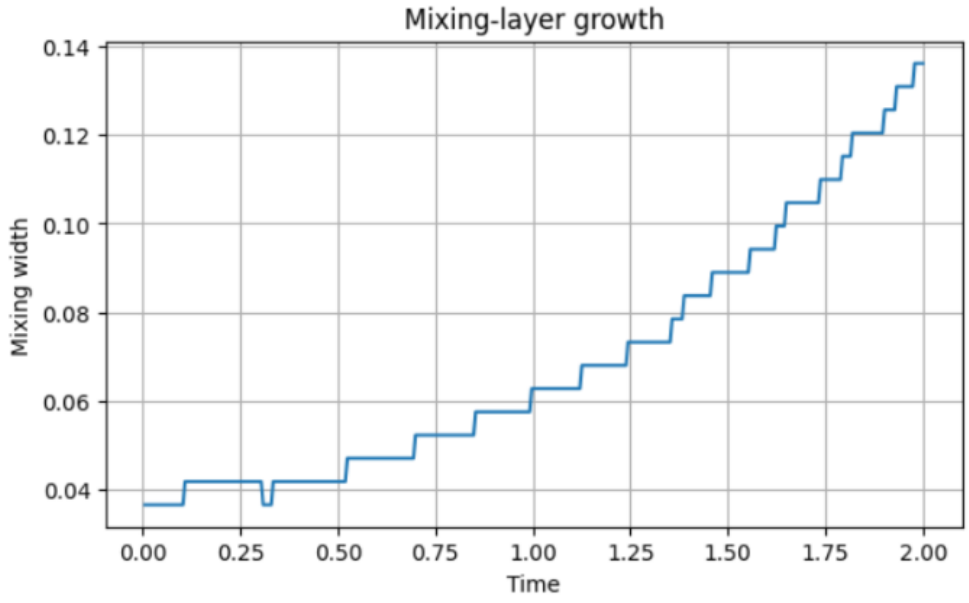
The A = 0.8 case develops a measurement artefact after approximately t = 1.5, so its later bubble and mixing-width values should not be interpreted as physical interface motion. The depletion of heavy-fluid fraction near the upper wall also indicates that further checks are needed despite the very small mass error. The source recommends examining velocity divergence, identifying spike and bubble locations from the alpha = 0.5 interface contour near the interface, and refining the grid.

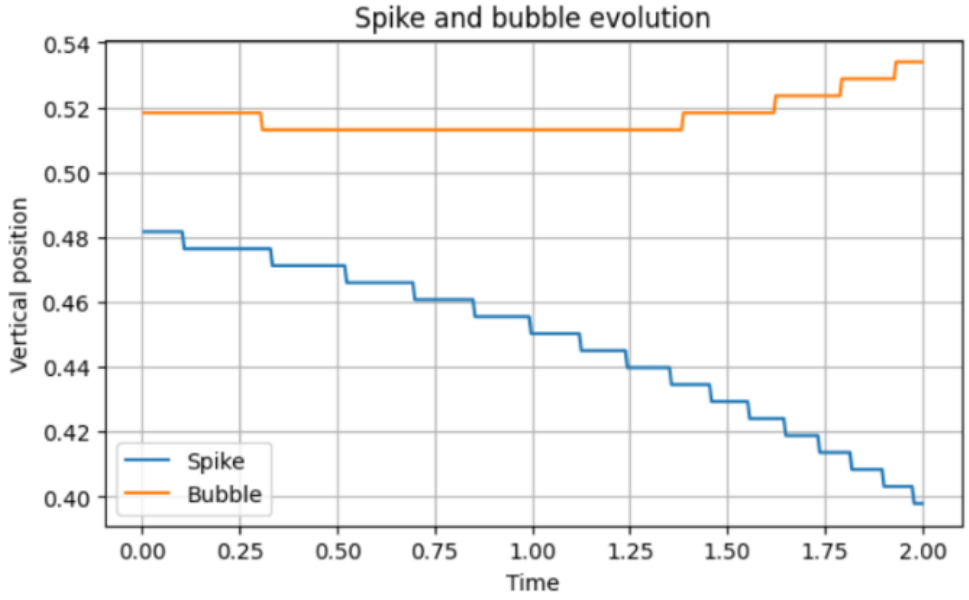
10. Original Simulation Figures

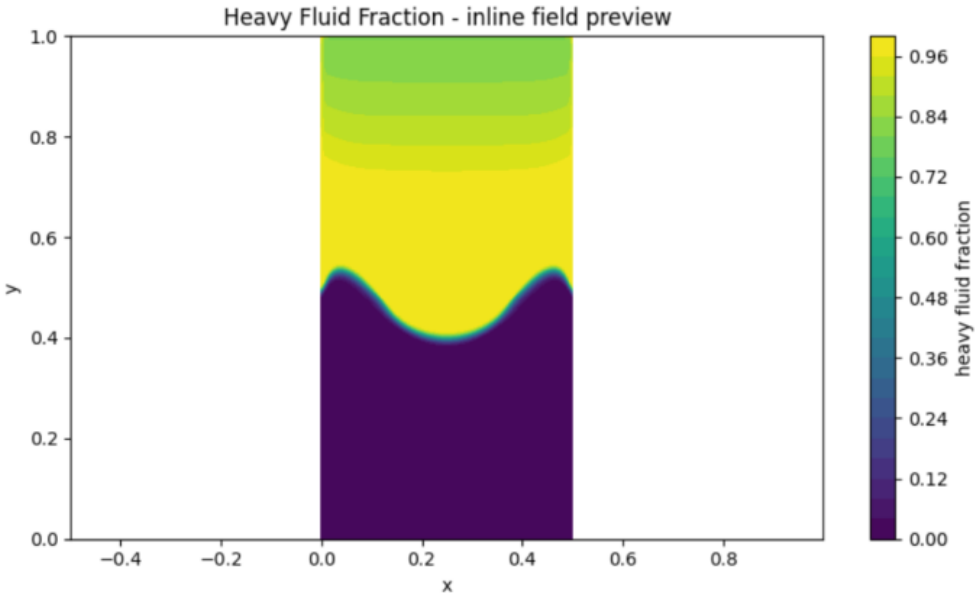
The original conservation histories, mixing-width curves, spike/bubble trajectories, phase-fraction contours and results table are retained in the following pages.

For Atwood number 0.2:

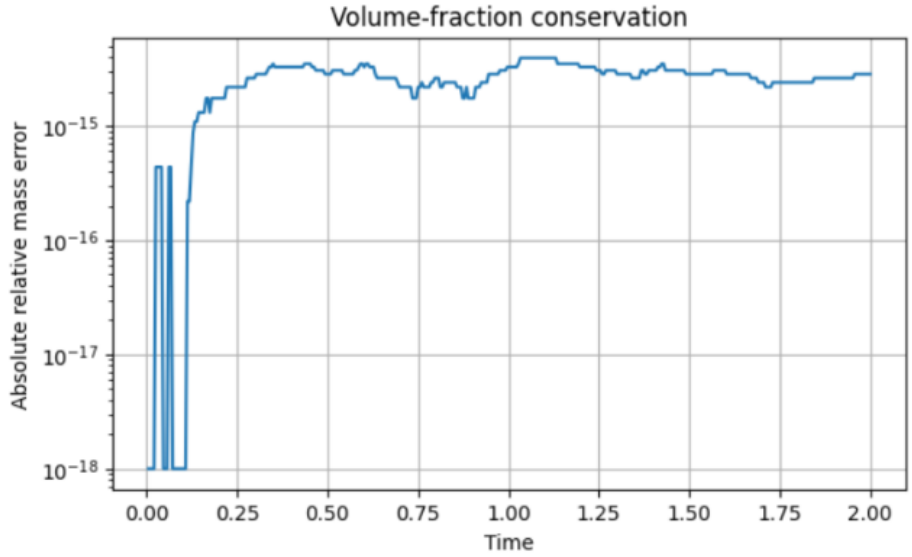


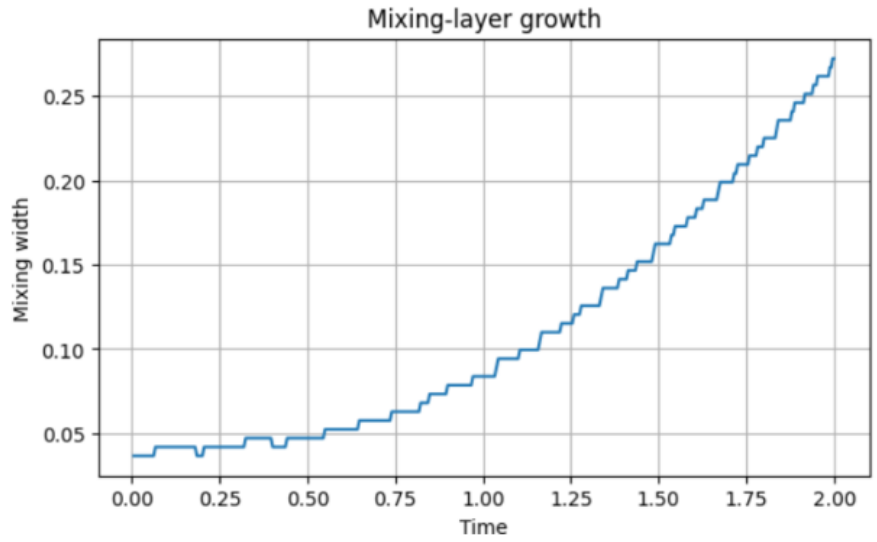


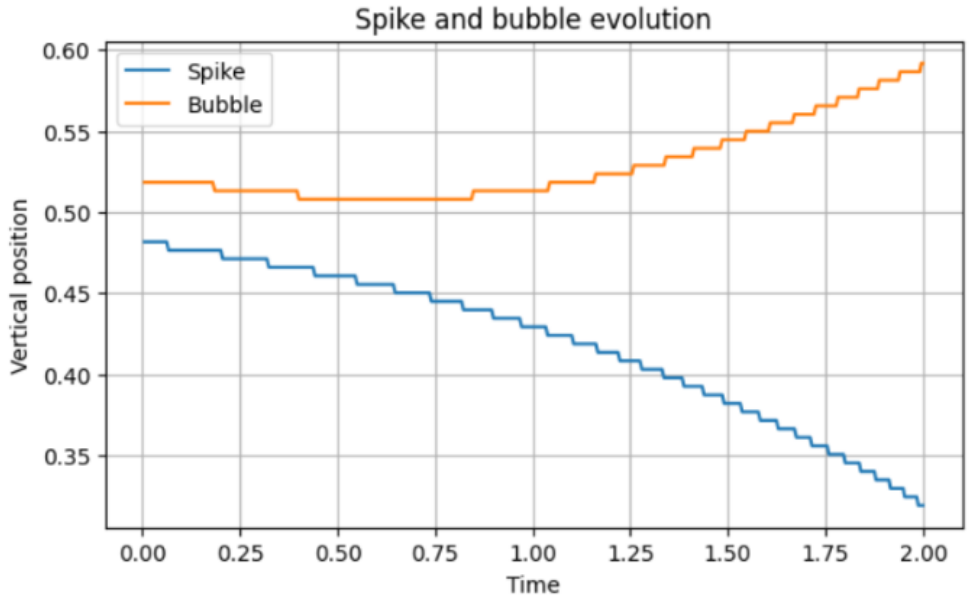

Spike and bubble evolution
Vertical position
Time
Spike
Bubble
0.54
0.52
0.50
0.48
0.46
0.44
0.42
0.40
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

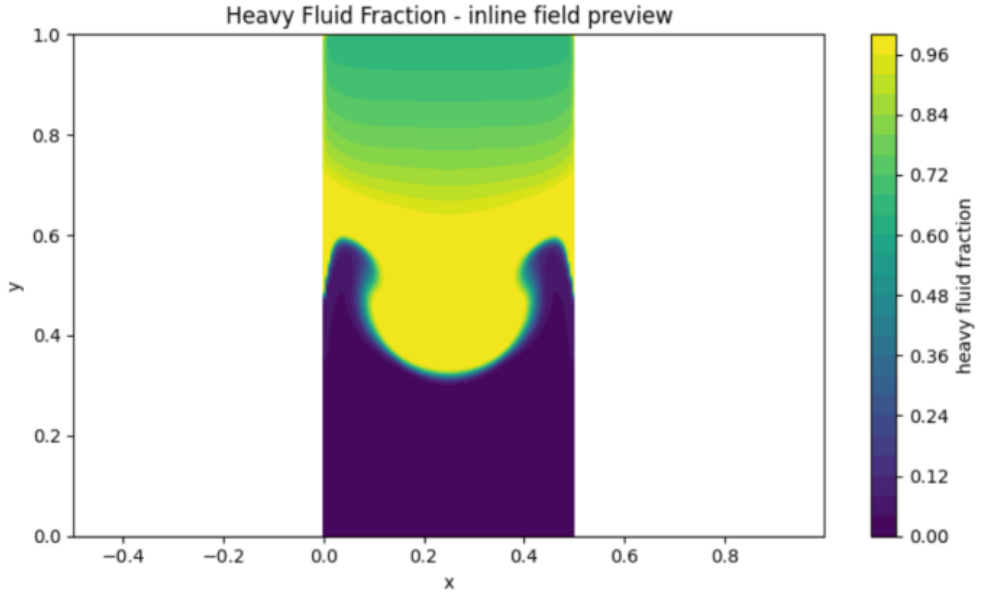

Heavy Fluid Fraction - inline field preview
y
x
heavy fluid fraction
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Atwood number 0.5:

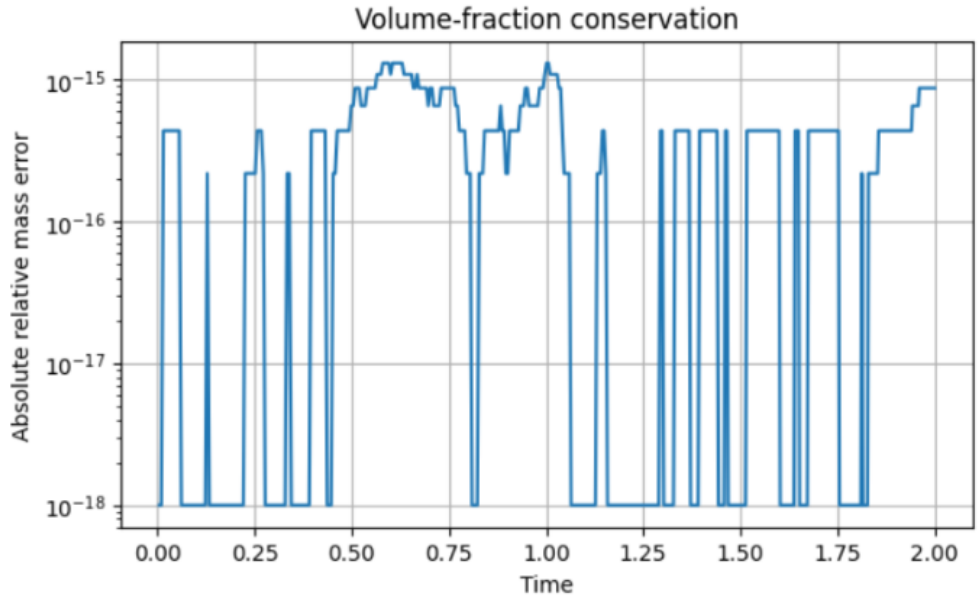


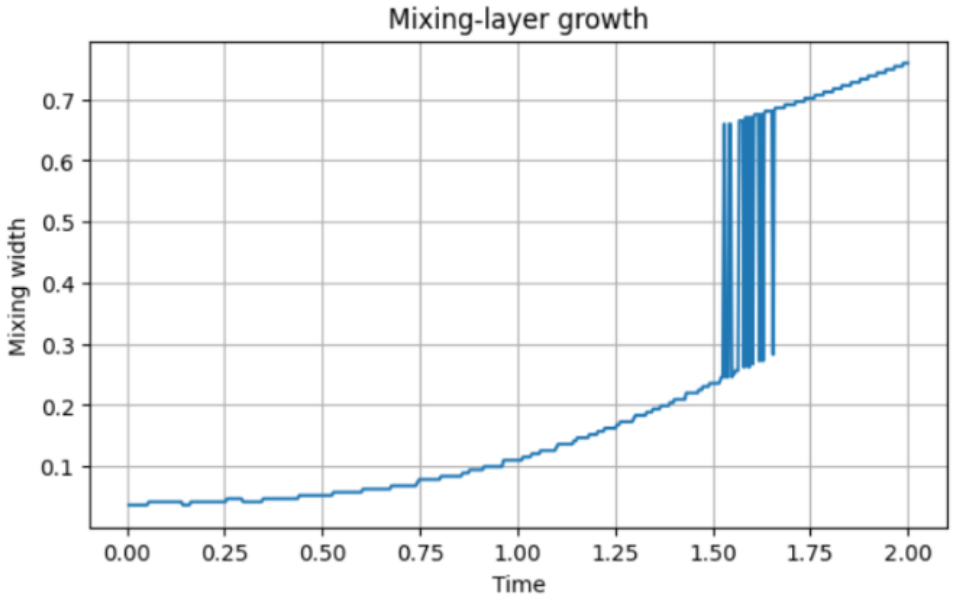


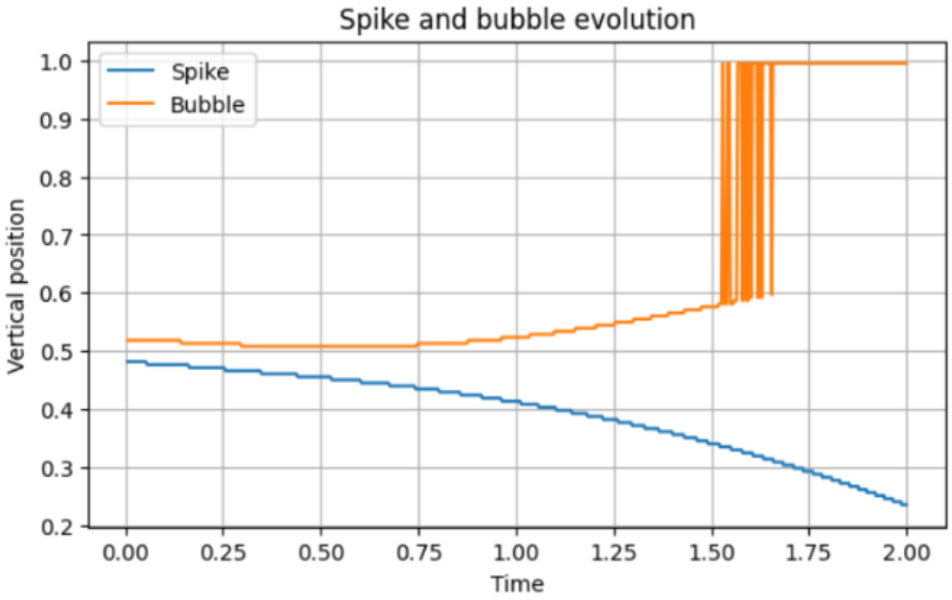

Spike and bubble evolution
Spike
Bubble
Vertical position
Time
0.60
0.55
0.50
0.45
0.40
0.35
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

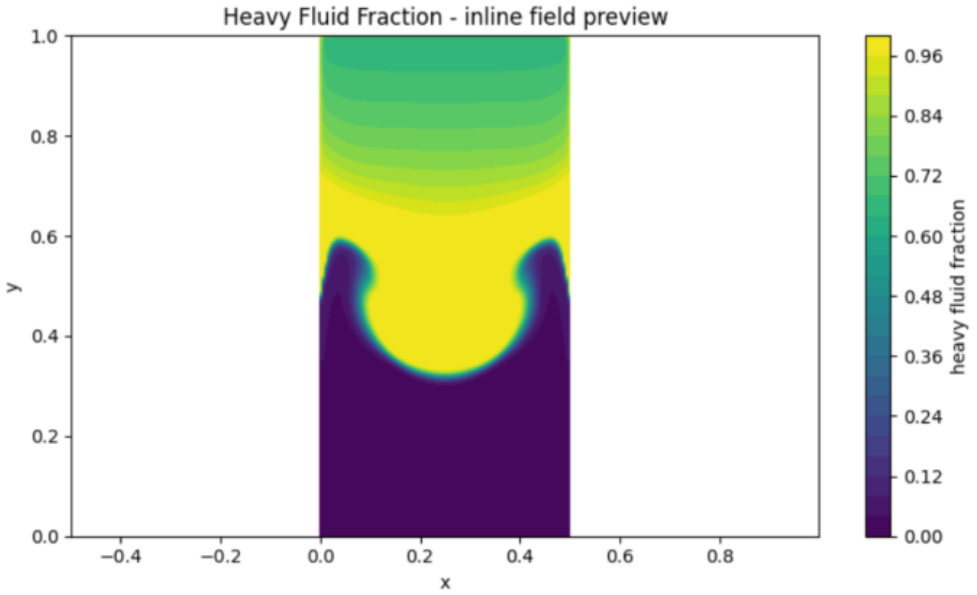

Heavy Fluid Fraction - inline field preview
heavy fluid fraction
x
y
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8

For Atwood number 0.8:






Spike and bubble evolution
Spike
Bubble
Vertical position
Time
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00


Heavy Fluid Fraction - inline field preview
y
x
heavy fluid fraction
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

Results:

Parameter	**A = 0.2**	**A = 0.5**	**A = 0.8**
Interface at t = 0 (spike / bubble position, y)	≈ 0.48 / 0.52	≈ 0.48 / 0.52	≈ 0.48 / 0.52
Spike position at t = 2 (y)	≈ 0.398	≈ 0.319	≈ 0.235
Spike descent from t = 0 to 2	≈ 0.084	≈ 0.163	≈ 0.247
Bubble position at t = 2 (y)	≈ 0.534	≈ 0.592	≈ 0.58 at t ≈ 1.5, then reads 1.0 (artefact)
Mixing width at t = 0	≈ 0.037	≈ 0.037	≈ 0.037
Mixing width at t = 2	≈ 0.136	≈ 0.272	≈ 0.76 (affected by artefact)
Net mixing-width growth	≈ 0.099 (at t = 2)	≈ 0.235 (at t = 2)	≈ 0.20 (at t = 1.5)
Maximum absolute relative mass error	$\approx 6 \times 10^{-15}$	$\approx 4 \times 10^{-15}$	$\approx 1.3 \times 10^{-15}$
Jumps in bubble position / mixing width	none	none	t ≈ 1.5–1.65
Phase-fraction range, $0 \le \alpha \le 1$ (field plot)	≈ 0 to 1	≈ 0 to 1	[INSERT: min/max α]
Heavy-fluid fraction near top wall at t = 2	≈ 0.8	≈ 0.6	[INSERT: from field plot]
Interface shape at t = 2	smooth, single spike at mid-width	rounded spike, curled tips at side walls	[INSERT: from field plot]

Conclusion:

The Rayleigh–Taylor instability of a single-mode perturbed heavy-over-light interface in a no-slip box was simulated using a projection flow solver with conservative upwind phase-fraction

transport, at Atwood numbers of 0.2, 0.5 and 0.8. In all cases the heavy fluid above penetrates downward as a spike at mid-width, while the lighter fluid rises along the side walls as bubbles. The interface starts at $y \approx 0.5$ with the spike at $y \approx 0.48$ and the bubble at $y \approx 0.52$. By $t = 2$ the spike has descended to $y \approx 0.398$, 0.319 and 0.235 for $A = 0.2$, 0.5 and 0.8 respectively, i.e. by about 0.084, 0.163 and 0.247. The bubble has risen to $y \approx 0.534$ for $A = 0.2$ and $y \approx 0.592$ for $A = 0.5$.

Effect of Atwood number on growth: Because the buoyancy term scales with the Atwood number, spike descent, bubble rise and mixing width all increase with A. The spike descent grows less than in proportion to A, in the ratio of about 1 : 1.9 : 2.9 for A in the ratio 1 : 2.5 : 4. The mixing width, which equals the bubble position minus the spike position in the plotted curves, grows from $\approx$ 0.037 at $t = 0$ to ≈ 0.136 ($A = 0.2$) and ≈ 0.272 ($A = 0.5$) at $t = 2$. The net growth of ≈ 0.099 and $\approx$ 0.235 is in the ratio of about 2.4, close to the Atwood-number ratio of 2.5. Growth is slow at first and accelerates: for $A = 0.2$ the net growth at $t = 2$ is about 3.8 times that at $t = 1$, close to quadratic in time, while for $A = 0.5$ the ratio is about 5. In every case the spike moves faster than the bubble, and the bubble lags most at $A = 0.2$. The bubble position first decreases slightly, by about 0.005–0.01, before it rises, which is consistent with the initial smoothing of the interface. The stair-step appearance of the curves reflects the discrete grid cells to which the positions are tied.

Interface shape: At $A = 0.2$ the interface remains smooth and close to a single-mode shape, with the spike at mid-width and the bubbles at the side walls. At $A = 0.5$ the spike has a rounded head, and the rising light fluid at the side walls develops curled tips, which indicates the onset of the nonlinear roll-up stage. This stage appears earlier and more strongly as the Atwood number increases.

Conservation and boundedness: The absolute relative mass error of the volume fraction stays at round-off level, below approximately 6×10^{-15}, 4×10^{-15} and 1.3×10^{-15} for $A = 0.2$, 0.5 and 0.8, because both the transport and the redistribution are conservative. The phase fraction stays within $0 \le \alpha \le 1$ in the field plots. However, conservation of the total phase fraction does not guarantee a physical field. In the field plots the heavy-fluid fraction in the upper layer decreases progressively towards the top wall, to about 0.8 for $A = 0.2$ and about 0.6 for $A = 0.5$ at $t = 2$, although no interface is present there. For an incompressible flow with an exactly divergence-free velocity this region should remain at $\alpha = 1$. This points to a residual velocity divergence acting through the conservative transport term, and/or to the clipping and redistribution step spreading phase fraction, so the boundedness is partly enforced by redistribution and not obtained from accurate transport alone.

Result at $A = 0.8$: The spike continues to descend to $y \approx 0.235$, but between $t \approx 1.5$ and 1.65 the bubble position repeatedly jumps from ≈ 0.6 to the top wall ($y = 1$), and the mixing width jumps from ≈ 0.27 to ≈ 0.7. After this the bubble remains at $y = 1$ and the mixing width reads ≈ 0.76 at $t = 2$, which equals $1.0 - 0.235$. Since the heavy-fluid fraction near the top wall is depleted more strongly as A increases, these jumps are most likely caused by the interface-detection criterion

picking up the diluted top layer, not by physical bubble rise. The spike, bubble and mixing-width values for A = 0.8 are therefore reliable only up to about t = 1.5.

Overall, the simulations reproduce the expected trend that Rayleigh–Taylor growth strengthens with the Atwood number, with a nearly proportional increase of the net mixing-width growth between A = 0.2 and 0.5, and conserve the phase fraction to round-off at every Atwood number. The depletion of the heavy-fluid fraction near the top wall and the artefact in the A = 0.8 interface metrics limit the quantitative reliability of the results. Checking the velocity divergence, defining the spike and bubble positions from the $\alpha = 0.5$ contour near the interface instead of from the global extent of the mixed region, and refining the grid would improve the quantitative reliability.